

are examples of precipitation.

pressure

The amount of force that is applied to a surface per unit of area.

protein

A type of complex chemical found in all living things, used as enzymes and in muscles. A protein is a chain of simple units called amino acids. There are about 20 natural amino acids, and a protein has hundreds of these units connected in a specific order.

protist

A single-celled organism with a complex cell structure, including organelles and a nucleus.

proton

A positively charged particle that is located in the nuclei of all atoms.

pupate

To prepare to change from a larva to an adult form (imago); for example, a caterpillar pupates as a chrysalis before emerging as a butterfly.

radiation

Waves of energy that travel through space. Radiation includes visible light, heat, X-rays, and radio waves; nuclear radiation includes subatomic particles and fragments of atoms.

radicals

Atoms, molecules, or ions with unpaired electrons that cause them to react easily.

radioactive

Relating to atoms that are unstable and break apart, releasing high-energy particles.

rarefaction

A decrease in the pressure and density of molecules along a longitudinal wave.

reactivity

A description of how likely a substance is to become involved in a chemical reaction.

reduction

When a substance gains electrons during a chemical reaction and so its oxidation number is reduced.

reflection

When a wave bounces off a surface.

refraction

When a wave changes direction as it passes from one medium to another.

repulsion

A force that pushes things apart; the opposite of attraction.

respiration

The process occurring in all living cells that releases energy from glucose to power life.

rubisco

Short for "ribulose biphosphate carboxylase oxidase," an enzyme that is responsible for taking carbon dioxide from the atmosphere and reacting it with water to make glucose as part of photosynthesis.

salt

An ionic compound formed by a reaction between an acid and base (including an alkali).

sedimentary rock

A rock that forms from sediments, which are layers of substances that have settled on the seabed or ground before becoming buried and compressed for millions of years.

solute

A substance that becomes dissolved in another.

solvent

A substance that can have other substances dissolved in it.

speed

The rate of how fast an object is moving.

states of matter

The three main forms of matter that a substance can take are: solid, liquid, or gas. Plasma is a fourth state of matter.

strain

The change of the shape of an object in response to stress.

stress

A force that alters the shape of an object, by stretching, bending, and sometimes breaking it.

subatomic particle

A particle that is smaller than an atom, such as a proton, neutron, and electron.

superconductor

A material that conducts electricity without warming up and so wastes none of the energy it is carrying.

suspension

A mixture in which small solids, blobs of liquid, or gas bubbles are spread throughout a liquid.

temperature

An average measure of the thermal energy or heat of an object.

torque

The turning effect of a force.

torsion

A twist caused by torque.

transformer

A device for altering the voltage of an electrical current.

transverse

A wave that moves by rising and falling perpendicular to the direction of its motion.

vapor

Another word for a gas.

vascular

Concerning vessels, tubes that transport substances around a body.

velocity

A speed of something in a particular direction.

vertebrate

An animal that has a vertebral column, a flexible spine made from a chain of smaller bones called vertebrae; the largest animals are vertebrates, and include fish, amphibians, reptiles, mammals, and birds.

vesicle

A membranous sac that contains a material being processed by a cell; a vesicle may be used to release substances from a cell.

voltage

A measure of the force that pushes electrons around an electric current.

xylem

The vascular tissue that transports water and minerals around a plant.

wavelength

The distance measured between any point on a wave and the equivalent point on the next wave.

weight

The force applied to a mass by gravity.

work

The amount of energy transferred when a force is being applied to a mass over a certain distance.